

HOMEWORK 5  
ELEMENTARY LINEAR ALGEBRA  
(MATH 2250-03), SPRING, 2018

CHECK: 03/15/2019

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SECTION 2.8 - SUBSPACES OF  $\mathbb{R}^n$

In problem 1 and problem 2, you are given a matrix  $A$  and an echelon form of  $A$ . Use this information to find a basis for  $\text{Col}(A)$  and  $\text{Nul}(A)$ .

1.  $A = \begin{bmatrix} 4 & 5 & 9 & -2 \\ 6 & 5 & 1 & 12 \\ 3 & 4 & 8 & -3 \end{bmatrix} \xrightarrow{\text{Row Reduce}} \begin{bmatrix} 1 & 2 & 6 & -5 \\ 0 & 1 & 5 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

2.  $A = \begin{bmatrix} 1 & 4 & 8 & -3 & -7 \\ -1 & 2 & 7 & 3 & 4 \\ -2 & 2 & 9 & 5 & 5 \\ 3 & 6 & 9 & -5 & -2 \end{bmatrix} \xrightarrow{\text{Row Reduce}} \begin{bmatrix} 1 & 4 & 8 & 0 & 5 \\ 0 & 2 & 5 & 0 & -1 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

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SECTION 2.9 - DIMENSION AND RANK

3. Let  $b_1 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$  and  $b_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ . Then,  $\mathcal{B} = \{b_1, b_2\}$  is a basis for the subspace of  $\mathbb{R}^2$  spanned by  $\mathcal{B}$ . Find the vector  $x$  determined by the  $\mathcal{B}$ -coordinate vector  $[x]_{\mathcal{B}} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$ .

4. Let  $\mathcal{B} = \{b_1, b_2\}$  with  $b_1 = \begin{bmatrix} -3 \\ 1 \\ -4 \end{bmatrix}$  and  $b_2 = \begin{bmatrix} 7 \\ 5 \\ -6 \end{bmatrix}$ . You may assume that  $x = \begin{bmatrix} 11 \\ 0 \\ 7 \end{bmatrix}$  is in the subspace of  $\mathbb{R}^3$  spanned by  $\mathcal{B}$ , and you may also assume that  $\mathcal{B}$  is a basis for this subspace. Find  $[x]_{\mathcal{B}}$ , the  $\mathcal{B}$ -coordinate vector of  $x$ .

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SECTION 3.1 - INTRODUCTION TO DETERMINANTS

5. Let  $A = \begin{bmatrix} 1 & -2 & 5 & 2 \\ 0 & 0 & 3 & 0 \\ 2 & -4 & -3 & 5 \\ 2 & 0 & 3 & 5 \end{bmatrix}$ . Compute  $\det(A)$ .

6. Let  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$  and let  $k \in \mathbb{R}$  be a scalar. Find a formula which relates  $\det(kA)$  to  $\det(A)$ .

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## SECTION 3.2 - PROPERTIES OF DETERMINANTS

7. Use determinants to decide if the set of vectors  $\left\{ \begin{bmatrix} 4 \\ 6 \\ 2 \end{bmatrix}, \begin{bmatrix} -7 \\ 0 \\ 7 \end{bmatrix}, \begin{bmatrix} -3 \\ -5 \\ -2 \end{bmatrix} \right\}$  is linearly independent.
8. Let  $A$  be an invertible  $n \times n$  matrix. Prove that  $\det(A^{-1}) = \frac{1}{\det(A)}$ .
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## SECTION 3.3 - CRAMER'S RULE, VOLUME, AND LINEAR TRANSFORMATIONS

9. Compute the adjugate of  $A = \begin{bmatrix} 1 & 1 & 3 \\ -2 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ . Then, using the adjugate, compute  $A^{-1}$ .
10. Let  $(x_1, y_1)$ ,  $(x_2, y_2)$  and  $(x_3, y_3)$  be points in  $\mathbb{R}^2$ , and let  $R$  be the triangular region in  $\mathbb{R}^2$  whose vertices correspond to these points. Write  $A(R)$  to denote the area of  $R$ . Show that

$$A(R) = \frac{1}{2} \det \left( \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} \right).$$